

Machine Learning for Canopy Classification

AGRON 5130 - Module 13 Assignment

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Introduction

The goal of this assignment is to interpret and reason through the machine learning workflow from Module 13.

You worked through a guided exercise where you:

- applied a simple threshold rule (ExG)
- evaluated model performance using confusion matrices
- trained a random forest model
- tested model generalization across dates

In this assignment, you will **analyze the results from that exercise** and explain what they mean.

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

Submission Notes

- Answers should be written in complete sentences
- Support your answers using results from the exercise (confusion matrices, maps, or observed patterns)
- Focus on reasoning and interpretation, not code

Part 1: Model Evaluation and Interpretation

Question 1

3 points

Using the confusion matrices from the exercise, compare the ExG threshold and the random forest model on the **second flight (test date)**. Describe the overall accuracy from each model (how many correct classifications and how many errors)

Question 2

3 points

The ExG threshold was chosen using the first flight image.

Compare its performance between:

- the first flight (training conditions)
- the second flight (new conditions)

1. Did performance remain stable across dates?
2. What does this tell you about using a **fixed threshold rule** in practice?

Question 3

3 points

The random forest model is more flexible than the ExG threshold rule. However, in this case, it did not dramatically improve performance.

Provide at least **one reason** why a more flexible model might fail to improve classification accuracy in this dataset. (Think about the data, not just the model.)

Question 4

3 points

Consider how classification errors affect agronomic interpretation. Think in terms of how canopy coverage is calculated. Your answer should connect model errors to their practical consequences.

Question 5

3 points

Propose **one specific improvement** to the classification workflow.

Your answer must include:

1. What you would change
2. Why you think it would improve performance
3. One potential limitation or tradeoff of your approach

(Examples: different features, different training data, different evaluation strategy. Be specific, please.)